

UNIT-3 | Lecture 4

Triggers & Embedded SQL

Coverage: Hour 6

Learning Objectives

Understand triggers, their events and timing.

Understand how embedded SQL connects application programs with databases.

Triggers

A **trigger** is a database program that automatically executes when a specified database event occurs.

Common events: **INSERT, UPDATE, DELETE**.

Common timing: **BEFORE** and **AFTER**; some systems also support **INSTEAD OF** triggers.

Uses of Triggers

Automatic auditing and logging.

Enforcing complex business rules.

Maintaining derived/summary information.

Recording important changes.

Preventing or controlling selected operations.

Conceptual Example

When a student's fee record is updated, a trigger can automatically store the student ID, old fee, new fee, and update time in an audit table.

Embedded SQL

Embedded SQL places SQL statements inside a host programming language/application.

The host program handles application logic while SQL performs database operations.

Host variables can pass values to SQL and receive query results, depending on the programming/database interface.

Flow

Application Program → SQL Statement → DBMS → Result → Application

Trigger vs Embedded SQL

Triggers	Embedded SQL
Automatic response to database events	Executed as part of an application
Stored/associated with database	SQL included in host program
Useful for rules and auditing	Useful for application-driven DB access

Key Takeaway

Triggers automate database actions; embedded SQL lets application programs interact with a DBMS using SQL.